PRADEEP RAJALU

India pradeeprajalu5@gmail.com 7010327639 linkedin.com/in/raju25

SUMMARY

Motivated AI/ML practitioner and Data Science with hands-on expertise in building end-to-end machine learning pipelines, preprocessing complex datasets, and designing predictive models using **Python, SQL, Scikit-Learn, and Power BI**. Well-versed in modern AI architectures, exploratory data analysis, and automation workflows. Eager to leverage technical rigor and practical engineering experience to build scalable, high-impact intelligent systems.

EXPERIENCE

AI Analyst Intern

IBM **August 2025 - August 2025, India**

- Engineered a real-time face detection and recognition system using Python, OpenCV, and deep learning models, achieving 95% identification accuracy and deploying via a user-friendly dashboard for security access control.
- Developed and tested predictive machine learning models (e.g., logistic regression, random forest) with scikit-learn to analyze customer churn patterns, directly informing retention strategies that reduced churn by 12%.
- Built modular Python scripts to automate data cleaning and feature engineering, cutting data preparation time by 40% for the analytics team.
- Created interactive Streamlit dashboards for real-time KPI visualization, enabling cross-functional stakeholders to self-serve insights without SQL or coding support.

DataScience

iStudio **June 2026 - Present, pune,**

- Developed and implemented end-to-end data science and machine learning pipelines using **Python**, leveraging libraries such as Pandas, NumPy, Scikit-Learn, and Matplotlib.
- Cleaned, preprocessed, and transformed large, unstructured datasets to ensure high data quality for modeling and exploratory data analysis (EDA).
- Built and optimized predictive models and classification/regression algorithms, evaluating performance metrics to improve model accuracy.
- Designed interactive data visualizations and dashboards using **Power BI** and **SQL** to communicate insights effectively to stakeholders.
- Collaborated with team members to solve complex business problems using data-driven approaches and modern data analytics workflows.

PROJECT

Ai Based journal paper writer

May 2026 - May 2026

- Used Python and a pre-trained language model (e.g., GPT-2 or T5) to generate short academic paragraphs from user-provided titles and keywords.
- Preprocessed a small dataset of 50–100 sample abstracts to fine-tune the model, improving relevance of generated text.
- Built a simple command-line or Jupyter Notebook interface to test and compare generated outputs manually.
- Documented the process and learned basic NLP techniques (tokenization, model inference, text cleanup).

face detection and face recognisation using python

March 2026 - May 2026

- Built a real-time face detection system using OpenCV Haar cascades, processing **15–20 frames per second** on a laptop webcam.
- Used a pre-trained `face_recognition` library to identify **up to 3 known individuals** from a small photo dataset (5 images per person).
- Achieved **~80% accuracy** in varied lighting conditions during a live demo with 10 test frames.

ICU rover ward

September 2025 - November 2025

- Assembled a Raspberry Pi-based rover with two DC motors and a webcam, controlled wirelessly via a Python script using keyboard commands.
- Achieved reliable movement and live video streaming within a **3-meter range** during indoor tests on tile flooring.
- Tested basic obstacle avoidance (manual stop within **30 cm**) using an ultrasonic sensor, demonstrating safe operation in narrow spaces.

Emotion recognisation using python

September 2025 - September 2025

- Built a real-time emotion recognition system using Python, OpenCV, and deep learning (CNN) to classify facial expressions into seven emotion categories (e.g., happy, sad, angry) with 88% accuracy.
- Preprocessed and augmented a dataset of over 35,000 facial images to improve model robustness across varying lighting conditions and angles.
- Deployed the model via a Flask web application, enabling live webcam-based emotion detection for user interaction testing.
- Optimized inference time by 30% through model quantization and batch processing, ensuring smooth real-time performance.

EDUCATION

BTech
Arunai engineering college • INDIA, Tiruvannamalai • 2021 - Now • 7.56

COURSEWORK

Prompt engineering
Infosys • 2025

Data visualization using python
GUVI • 2025 • Python

AWS AI Partition
AWS • 2025

Generative AI
GOOGLE • 2025

SKILLS

Programming & Scripting: Python, R, SQL

Machine Learning & AI: scikit-learn, PyTorch, TensorFlow basics

Data Science Tools: Pandas, NumPy, Matplotlib, Seaborn

Deployment & Visualization: Streamlit, Power BI, Tableau (basic)

Version Control & Collaboration: Git/GitHub

Big Data & Cloud: Hadoop/Spark basics, Google Colab, AWS (beginner exposure)

Languages: English, Tamil, Japanese

LLMs & Generative AI: GPT/LLaMA-family fine-tuning, LoRA, QLoRA, RLHF, Prompt Engineering, RAG, LangChain, Hugging Face Transformers, Vector Databases (Pinecone, FAISS, Weaviate)

MLOps & Deployment: Docker, Kubernetes, MLflow, Kubeflow, Airflow, Model Monitoring & Drift Detection